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Gravity constraints on the structure and volcanic evolution of Tenerife, Canary Islands

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Abstract. A detailed gravity study of Tenerife constrains the internal structure of the island and its active magma system. The maximum Bouguer anomaly (2636 gu, 1 gu = 10^{-6} m s^{-2} = 0.1 mGal) occurs over the SW part of the postshield Las Cañadas edifice (LCE), while smaller highs occur over the eroded shield volcanoes. The main high extends NE and NW over the adjacent Dorsal and Santiago rift zones. A deep low occurs over the northern LCE and Teide volcano. The power spectrum of the anomalies suggests shallow causative structures (< ~3 km), inferred to belong to the island's dense core. A model computed from the anomalies uses core and carapace densities of 2900 and 2360 kg m^{-3}, respectively, except beneath the main high, where the core requires a higher density (3210 kg m^{-3}). This part of the body is interpreted as a buried mafic volcano of substantial size, the Boca Tauce volcano, containing an old mafic plutonic complex. The enigmatic Las Cañadas caldera that truncates the LCE shows strong asymmetry. Dense materials occupy the southern part whereas low-density materials occupy the northern part to 2-3 km depth, consistent with a vertical collapse origin. High-density materials in the south are related to the intracaldera extension of the Dorsal rift zone. The boundary between subcaldera domains is interpreted as the buried head of the giant Icod landslide valley. The Icod landslide appears to have modified the caldera, truncating its northern side, and to have subsequently controlled the location of the active Pico Teide-Pico Viejo vent systems. Voluminous chaotic deposits lying off northern Tenerife, previously associated with the formation of the caldera and Icod valley, are better interpreted as being derived from older mass wasting events. The southern LCE has been buttressed against lateral failure by the buried Boca Tauce volcano.

1. Introduction

Tenerife, culminating in the active stratovolcano Teide (3718 m), is the largest of the Canary Islands (Figure 1). Teide is the focus of a long-lived, postshield magma system that has fed multiple cycles of basanite-phonolite activity from a large, central complex and persistent mafic activity along prominent rift zones. Historic eruptions on Tenerife have occurred mainly from the rifts and have been relatively infrequent (~1 per 100 years) compared to similarly sized, essentially basaltic oceanic volcanoes. This can be attributed to the central magma system having been occupied by felsic magmas since ~15 ka [*Ablay et al.*, 1998]. However, despite potential hazards associated with the active system, little is known in detail about the internal structure of the island, which might help to understand its structure, mechanics, or evolution and to constrain its possible future behavior. The absence of ongoing eruptions and associated seismicity has precluded geophysical investigations of magma movements, which have yielded important insights into basaltic systems such as those of Hawaii. Understanding the Tenerife system requires detailed geological, petrological, and passive geophysical study. Here we present the results of a new gravity study of Tenerife which builds on previous surveys [*Vieira et al.*, 1986; *Camacho et al.*, 1991]. Three-dimensional modeling of the observed gravity anomalies places constraints on the composition and distribution of subsurface lithologies, while interpretation of the gravity models in light of recent seismic, magnetic, and petrological studies [*Watts et al.*, 1997; *Blanco*, 1997; *Ablay et al.*, 1998], together with comparisons with other ocean island volcanoes [e.g., *Zucca et al.*, 1982], provides insights into the internal structure and postshield evolution of the island and magma system.

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2. Geological Background

The subaerial development of Tenerife began in the late Miocene with the growth of multiple mafic shield volcanoes now preserved as eroded massifs in the extremities of the island (Figure 1). K-Ar data [*Abdel-Monem et al.*, 1972; *Ancochea et al.*, 1990] suggest that the shield edifices developed and coalesced mainly between 6.0 and 4.5 Ma, in accord with an age of 6 Ma estimated from offshore seismic reflection data [*Funck et al*, 1996]. K-Ar dates of 8.0-16.1 Ma for shield rocks possibly reflect sample alteration or the presence of excess Ar [*Mitjavila and Villa*, 1993].

Overlying the shield is an extensive postshield complex, the main constituent of which is the Las Cañadas edifice (LCE). The

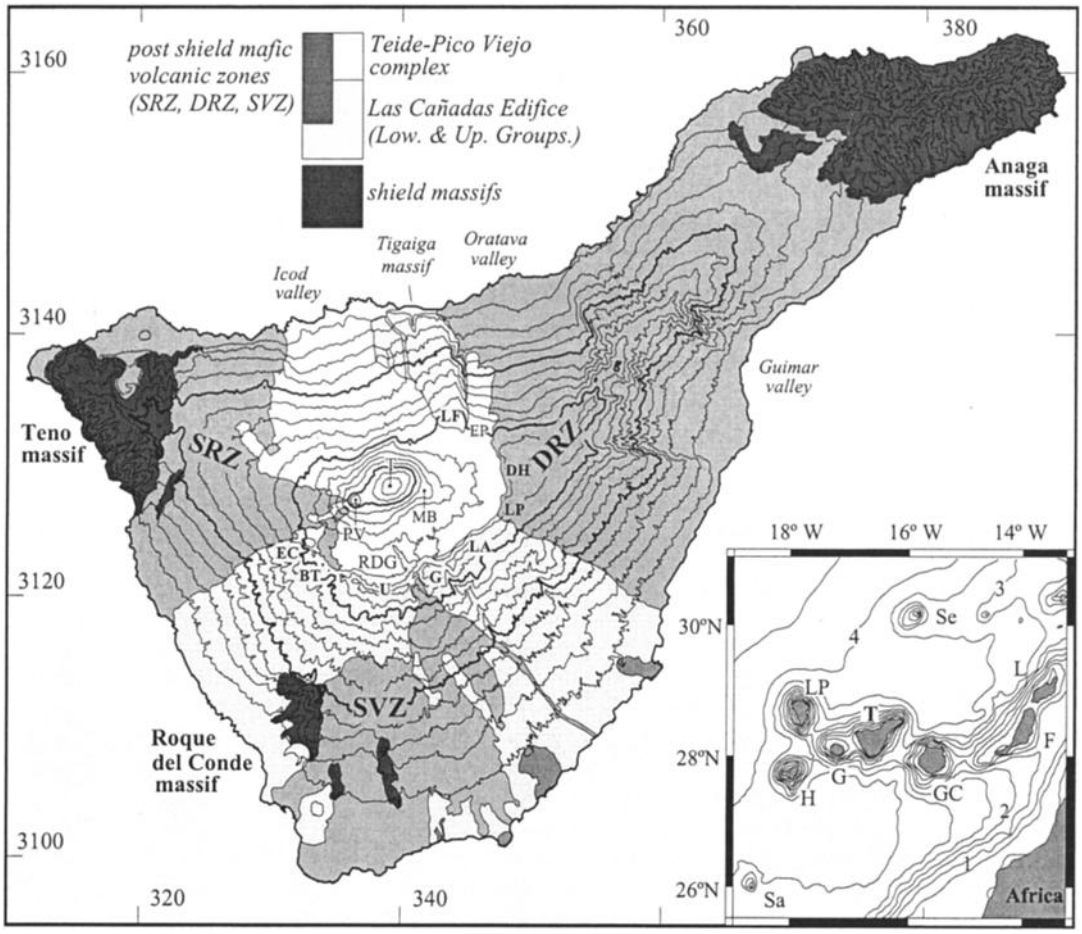

Figure 1. Simplified geological and topographic map of Tenerife showing the distribution of the main volcanic formations. Main map shows Las Cañadas caldera wall localities (EC, El Cedro, BT, Boca Tauce, U, Ucanca, RDG, Roques de Garcia, G, Guajara, LA, Las Angosturas, LP, Las Pilas, DH, Diego Hernández, EP, El Portillo, LF, La Fortaleza), Teide-Pico Viejo complex, (T, Teide volcano, PV, Pico Viejo volcano, MB, Montaña Blanca), and post-shield mafic volcanic zones (SRZ, Santiago rift zone, DRZ, Dorsal rift zone, SVZ, Southern volcanic zone). Names and locations of Upper Group landslide valleys are also shown. Contour interval is 200 m. Coordinates refer to 20 km squares of the Spanish national grid (Universal Transverse Mercator, UTM). Inset map shows the location and distribution of the Canary Islands. Main islands (LP, La Palma, H, El Hierro, G, La Gomera, T, Tenerife, GC, Gran Canaria, F, Fuertaventura, L, Lanzarote), and islets and seamounts (Sa, Saharan seamounts, Se, Selvagem islands). Bathymetric contours in kilometers below sea level after *Watts et al.* [1997].

LCE is elongated NE-SW and comprises several overlapping centres [*Fuster et al.*, 1968; *Araña*, 1971; *Martí et al.*, 1994; *Ancochea et al.*, 1990, 1995]. The SW part is the oldest, dating back to at least 3.5 Ma [*Huertas et al.*, 1994; *Fuster et al.*, 1994]. Chronostratigraphic work on the LCE by *Martí et al.* [1994] has distinguished a dominantly mafic Lower Group (>3.5-2.1 Ma) and an Upper Group comprising three formations of felsic volcanics; Ucanca (1.59-1.18 Ma), Guajara (850-570 ka), and Diego Hernández (370-179 ka) (Figure 2). These are thick sequences of phonolite lavas and pyroclastics, the latter correlating distally with a thick pyroclastic and volcaniclastic apron covering the south of the island, the Bandas del Sur [*Walker*, 1981; *Teide Group*, 1997; *Bryan et al.*, 1998]. The distribution of pyroclastic rocks suggests that the focus of Upper Group activity migrated NE with time along the summit of the LCE [*Araña*, 1971; *Martí et al.*, 1994; *Bryan et al.*, 1998].

The LCE is truncated by the Las Cañadas caldera, which is bounded to the south and east by high walls (Figure 1). Roques de García, a spur of Lower Group material abutting the south wall, divides the caldera into two unequal sectors. The northern and western caldera walls are absent except for La Fortaleza, an isolated segment forming part of the Tigaiga massif [*Ibarrola et al.*, 1993]. This separates the Icod and Oratava valleys which are scars interpreted to have resulted from major sector collapses [*Bravo*, 1962; *Coello*, 1973; *Navarro and Coello*, 1989]. The Las Cañadas caldera is a multiple depression formed during Upper Group times [*Araña*, 1971]. The ending of the Upper Group felsic cycles with voluminous (5-15 km^3) pyroclastic eruptions, subsequent lengthy periods of dominantly mafic volcanism and the migration of felsic volcanism between cycles, are consistent with a series of vertical caldera collapse events destroying shallow, differentiated magma systems [*Martí et al.*, 1994]. However, the absence of the northern caldera walls and the presence of inferred landslide scars suggest that the caldera was originated by giant, northward directed landslides [*Coello and Bravo*, 1989; *Navarro and Coello*, 1989; *Ancochea et al.*, 1995].

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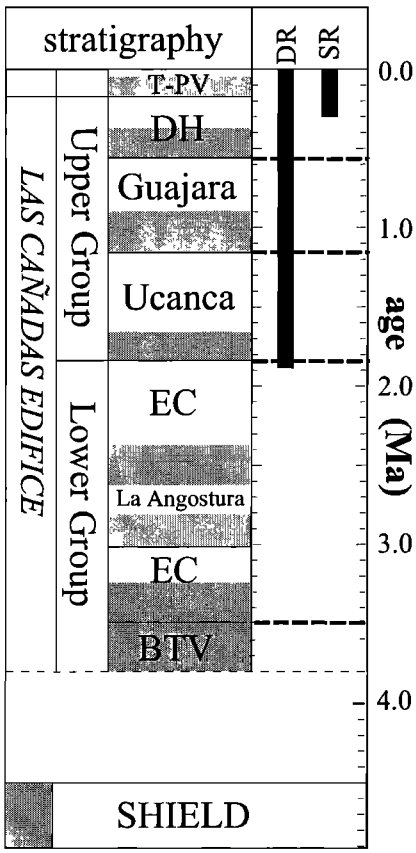

Figure 2. Summary volcanic stratigraphy for Tenerife. (left) General stratigraphy of the postshield central volcanic complex. The Las Cañadas edifice is divided into Lower and Upper Groups after *Martí et al.* [1994]. Dark and light shading indicate periods dominanted by mafic/intermediate and felsic volcanism, respectively. The Upper Group comprises three formations each exhibiting a transition from mafic/intermediate to felsic (see text). The Lower Group includes three main groups of felsic rocks and the products of the Boca Tauce volcano (see text). Age data for central volcanic complex (all K-Ar unless otherwise specified) are from *Ancochea et al.* [1990, 1995], *Mitjavila and Villa* [1993] (Ar-Ar), *Martí et al.* [1994], *J. Martí*, (unpublished data, 1996), *Huertas et al.* [1994], and *Fúster et al.* [1994]. (right) Rift volcanism and caldera collapse phases. Vertical bars indicate rift zone activity: DR, Dorsal rift; and SR, Santiago rift. Age data for rifts are paleomagnetic data for historic and subhistoric basalts [*Soler and Carrecedo,* 1986], and K-Ar data for Dorsal Ridge dikes [*Feraud et al.*, 1985]. Thick dashed lines indicate inferred caldera collapse events terminating the Upper Group felsic volcanic cycles [*Martí et al.*, 1994]. Approximate limit of shield volcanism is after *Abdel-Monem et al.* [1972] and *Ancochea et al.* [1990]. A possible period of volcanic quiescence or erosion occurring between the shield and post-shield stages is indicated.

This is apparently supported by a voluminous submarine apron of debris avalanche deposits lying off the island's north flank [*Watts and Masson*, 1995]. There is a correspondence between the inferred ages of the Icod (~0.17 Ma) and Oratava (<0.73 Ma) valleys and the ending of the Guajara and Diego Hernàndez cycles at ~0.18 Ma [*Mitjavila and Villa*, 1993] and ~0.56 Ma (Figure 2). This has been used to suggest that lateral collapses may have occurred in response to caldera collapse events [*Martí et al.*, 1997].

Volcanism since the last caldera-forming event has constructed the Teide-Pico Viejo complex, which occupies and spills out of the Las Cañadas caldera (Figure 1). It comprises intermediate to felsic lava sequences erupted from the stratocones Teide and Pico Viejo, overlying a succession of mafic volcanics. Teide-Pico Viejo products have become more differentiated with time and comprise two magmatic lineages; one erupted from near the base of the crust and another that differentiated extensively within a crustal magma chamber [*Ablay et al.*, 1998]. Teide and Pico Viejo have both experienced summit collapses associated with young phonolite eruptions, most recently at ~2 ka when a sub-plinian eruption from Montaña Blanca, the main satellite vent of the complex, triggered collapse of Pico Viejo, modifying a preexisting summit caldera [*Ablay et al.*, 1995].

Postshield volcanism has also occurred to the NW and NE of the LCE along prominent rift zones (Figure 1). The rifts are characterized by monogenetic cones built on subparallel mafic fissure systems concentrated near the crests of the steep-sided Dorsal and Santiago Ridges. The Dorsal rift has been active since Ucanca times (Figure 2), whereas the age of the Santiago rift is poorly constrained. Both have been active during Teide-Pico Viejo to Recent times, forming an axis of extension, intrusion, and eruption connecting the Santiago rift, through the intracaldera Teide-Pico Viejo vent systems, to the Dorsal rift. Flank volcanism has also occurred on the south side of the LCE in what is termed the Southern volcanic zone (Figure 1). This is a broad zone of mafic fissure vent systems on the lower to middle flank of the LCE rather than a well-defined rift zone [*Martí et al.*, 1996; *Bryan et al.*, 1998].

3. Gravity study

A number of gravity surveys have been undertaken on ocean island volcanoes, including those of the Hawaiian chain [*Kinoshita*, 1965; *Kinoshita and Okamura*, 1965; *Strange et al.*, 1965; *Zucca et al.*, 1982], La Réunion [*Rousset et al.*, 1989], and Iceland [e.g., *Gudmunsson and Milsom*, 1997]. The first interpretation of gravity anomalies in Tenerife was by *MacFarlane and Ridley* [1968], whose gravity anomaly map, though based on relatively few measurements, was able to identify a gravity maximum situated to the south of Teide and suggested the presence of NW-SE and NE-SW trending gravity ridges. Their preferred interpretive model assumed a dense intrusive body of conical form extending from the Moho to within 4 km of the surface whose density increased with depth from 2800 to 3100 kg m^{-3}. This was interpreted as representing lithological variation in the anomalous body due to the effects of magmatic differentiation. *Dash and Bosshard* [1969] and *Bosshard and MacFarlane* [1970] extended the gravity survey to marine areas around the Canary Islands. The latter authors produced a similar, two-dimensional model for Tenerife, comprising a single body with a density of 3100 kg m^{-3} extending to the Moho, which was interpreted as a subvolcanic intrusion. Work undertaken to define more precisely the gravity field of Tenerife was presented by *Vieira et al.* [1986], who described new measurements in the Las Cañadas caldera, and by *Camacho et al.* [1991], who augmented these data and produced a three-dimensional model of the subcaldera structure. This study demonstrated the presence of a low-density caldera fill, aligned NE-SW and extending to some 3 km below sea level, surrounded to the west, south and NE by higher-density features. High-

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density features also occurred at greater depths to the south and SE of the Las Cañadas caldera and beneath its eastern wall.

Watts et al. [1997] provided an interpretive two-dimensional model for the flanks of Tenerife and adjacent ocean crust based on offshore seismic reflection and gravity profiles, seismic refraction data, and assumed relationships between seismic velocity and density. Their NE-SW cross section showed a volcanic core of density 2720 kg m^{-3} overlain by a carapace comprising a lower layer of density 2480 kg m^{-3} and an upper layer of density 2270 kg m^{-3}. The carapace thickness was interpreted to be much greater over the north part of the island than the south. *Watts et al.* [1997] also identified a flexural moat around the island, occupied by unconsolidated sediments (2080 kg m^{-3}) and proposed that the oceanic crust and overlying sedimentary cover beneath Tenerife has been depressed 2-3 km by the load of the island.

3.1. Gravity Survey and Reduction

The present study is based on 971 gravity measurements on Tenerife, including those of previous surveys [*Vieira et al.*, 1986; *Camacho et al.*, 1991]. Observations were made with a LaCoste-Romberg gravimeter. Station locations (see Figure 5) were established from maps and by the use of differential Global Positioning System (GPS). Station elevations were generally determined by barometric altimetry, although some were obtained by optical leveling. Reduction to simple Bouguer anomalies was achieved using standard methods [*Kearey and Brooks*, 1991]. A constant regional gravity field was assumed, based on available marine data (J. Dañobetia, unpublished data compilation, 1999).

On a mature, active, volcanic island such as Tenerife, where the topography is extreme and elevation varies in excess of 3.7 km above and below sea level, terrain effects are large and show considerable spatial variations. The approach adopted in the computation of terrain corrections was as follows. A set of digitized elevation and bathymetric data for the Tenerife region was obtained (R. Vieirra, unpublished data, 1999) comprising 8462 points (Figure 3). In the topographic data set the density of digitization increases from marine areas to the Las Cañadas area, as the topography becomes more varied and complex. Digitization intervals in the latter are of the order of 200 m. Data were interpolated onto regular grids. In marine areas, the grid interval was 3 km; in marine areas around the island and land areas distant from the caldera, the grid interval was 1 km; in the Las Cañadas area the grid interval was 200 m. The terrain correction for each gravity observation was then computed using the gridded topography and assuming that it had the form of a suite of right-rectangular prisms, each of which was centered on a grid node and with a length equal to the relative elevation or water depth. The maximum value of the terrain correction so derived was 1086 gu (1 gu = 10^{-6} m s^{-2} = 0.1 mGal), situated at [339 3129] at the summit of Teide. Because of the uncertainties in determining the location, elevation, and terrain effects at each station, it is estimated that the error in the full Bouguer anomalies is of the order of ±50 gu.

Selection of an appropriate density for the Bouguer and terrain corrections is problematic because of the variability in density of the lithologies present. The approach adopted in selecting this density makes use of an extension to three dimensions of Nettleton's method [*Kearey and Brooks*, 1991]. Bouguer anomalies were computed for a range of densities for the Bouguer and terrain corrections. For each density the correlation coefficient between Bouguer anomaly and elevation was determined. The density that provides the least correlation should then represent a reasonable mean density of the surficial materials. A graph of correlation coefficient against reduction density for all observations on the island (Figure 4, curve a) reveals that a density of 2460 kg m^{-3} produces the minimum correlation, and this value was adopted in the reduction.

3.2. Gravity Anomaly Map

A contour map of the fully corrected Bouguer anomalies (Figure 5) shows that the maximum anomaly of 2636 gu occurs at Boca Tauce [333 3121], outside the SW wall of the Las Cañadas caldera. The gravity maximum lies at the center of a subcircular

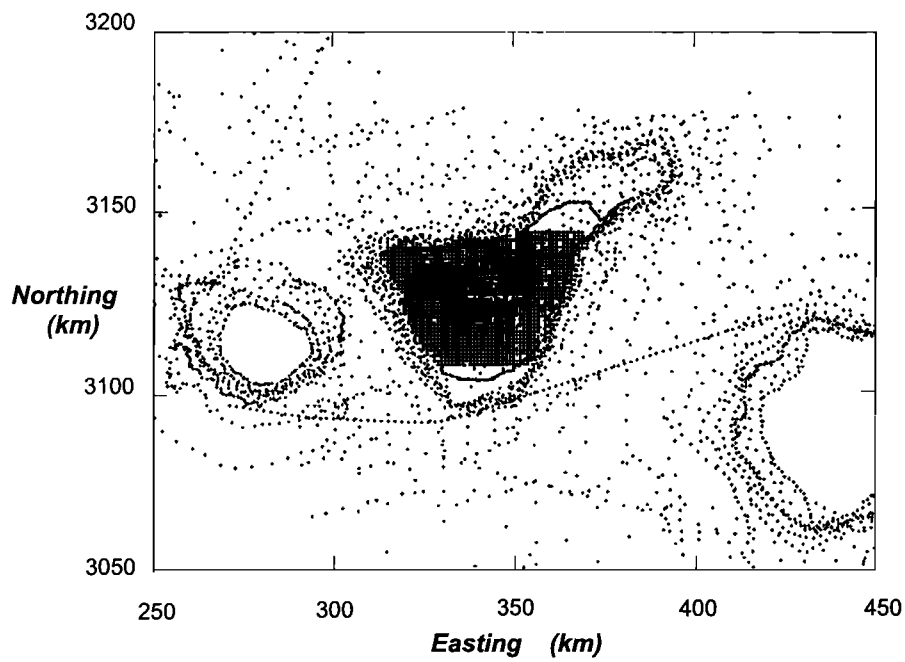

Figure 3. Locations of digitized heights and depths. The axes correspond to the Spanish national grid (km UTM).

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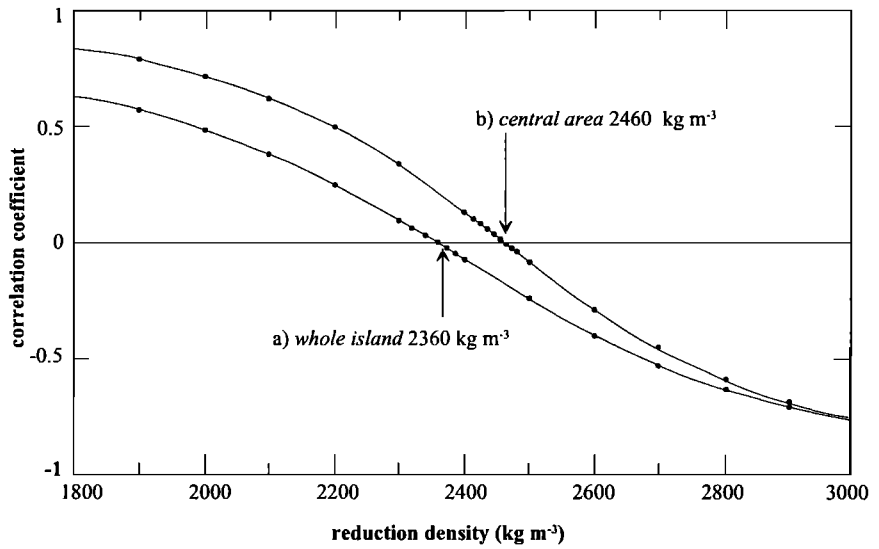

Figure 4. Graphs of correlation coefficients between topography and terrain-corrected Bouguer anomalies against reduction density for, curve a, the whole island, and, curve b, the central region of Tenerife containing the Las Cañadas caldera. The lower reduction density deduced for the central region is attributed to the accumulation of low-density, differentiated volcanics around the central complex.

gravity high located over the western central part of the island. This gravity high correlates with a negative magnetic anomaly of high amplitude [*Blanco*, 1997] and with a zone of high seismic velocity (P wave velocity [V_p] 5-6 km s^{-1}) that forms the core of the island [*Watts et al.*, 1997]. The main gravity high extends to the NE as a gravity ridge that follows the long axis of the LCE and Las Cañadas caldera and the topographic expression of the Dorsal Ridge. Two smaller gravity highs are located over the Anaga shield massif. The main gravity high also connects to the NW with a strong gravity high over the Teno massif. To the south and east, the main gravity high is bounded by a strong gravity gradient of some 80 gu km^{-1}, approximately normal to the SE

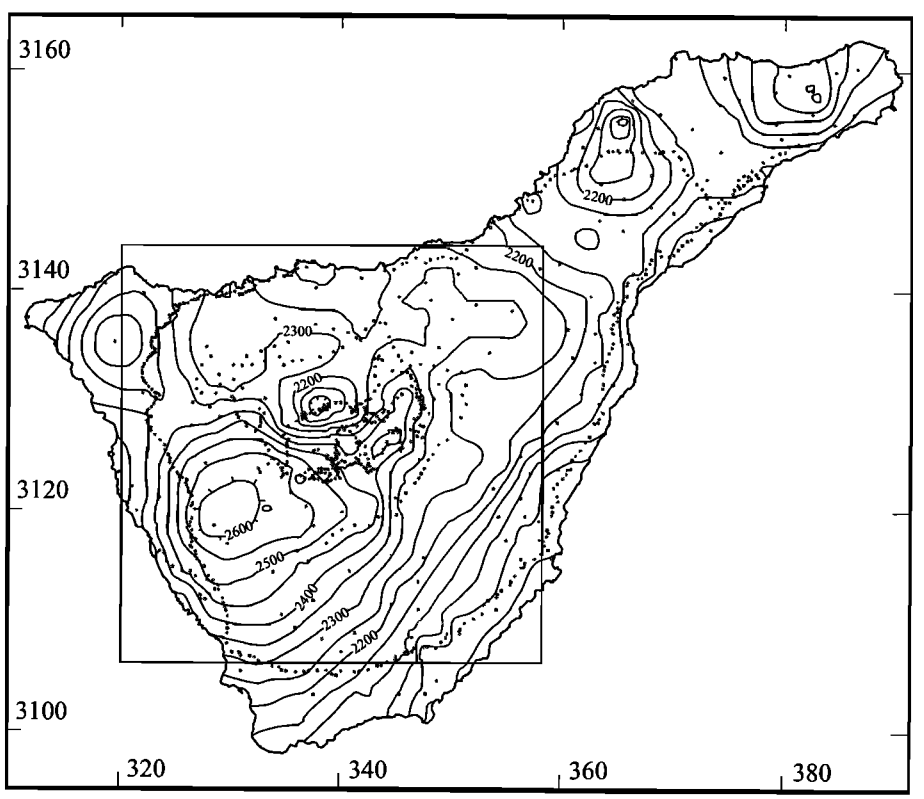

Figure 5. Bouguer anomaly map of Tenerife. Shaded circles indicate gravity stations. Contour interval is 50 gu. The square shows the area covered by Figures 6 and 7. The axes correspond to the Spanish national grid (km UTM east and north).

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coast. A much lower gravity gradient characterizes the northern boundary of the main high. Within the Las Cañadas area, there is a strong gravity low over the northern part of the Las Cañadas caldera and Teide edifice.

3.3. Gravity Modeling

Central Tenerife was selected for gravity modeling. An isogal map of the central area was constructed using a reduction density of 2360 kg m^{-3}, derived from a similar extended Nettleton's method as that described above (Figure 6). Gravity anomalies were modeled using an extension of the method of *Cordell and Henderson* [1968], in which the digitized gravity anomaly of the anomalous body is simulated by a suite of right-rectangular prisms, one per digitized anomaly point, each with a 2 km x 2 km horizontal section and a specified, variable density, whose vertical extent is automatically adjusted until the calculated anomaly matches the observed. The levels of either the tops or bases of the models must be specified. Owing to the inherent ambiguity in the interpretation of gravity anomalies, no unique model solution can be derived. Both types of model were constructed to provide possible end-members of the range of putative causative structures. In both models the putative structures are represented by contours drawn on the centers of either the tops or the bases of the model prisms. Model densities were selected on the basis of new measurements of the densities of volcanic lithologies on Tenerife (Table 1) and estimates of the densities of related plutonic cumulates (Table 2) and were varied between models in order to optimize the model fits. Visual inspection of the observed and calculated anomaly maps and profiles was used to judge the goodness of fit.

3.3.1. Model 1. The first model (Figure 7a) assumes a causative body with a fixed top lying at a depth of 50 m below the ground surface. Calculated anomalies are shown in Figure 7b. Best fit density contrasts were as follows: The part of the body lying beneath the central gravity high was given a density contrast of 840 kg m^{-3} (corresponding to a density of 3200 kg m^{-3}), the part under the gravity low over the Teide edifice 190 kg m^{-3} (2550 kg m^{-3}), and the rest of the area 640 kg m^{-3} (3000 kg m^{-3}). The modeled body is sufficiently thin beneath Teide that the assumed density contrast is not critical to the interpretation. The amplitude of the central positive anomaly is barely simulated; the model requires deep (5-6 km) downward protrusions in this area. The depths to the bases of these features are not well constrained as, beyond a certain depth, a large increase in column length has only a small effect on the surface gravity anomaly. In this model the gravity low over Teide volcano appears not to be caused by materials of particularly low density, but to be a result of the combined effects of the decaying positive anomalies associated with parts of the causative body to the SW and NE. In the model a NE-SW trending offshoot of the main body underlies the western Dorsal Ridge.

3.3.2. Model 2. The second model (Figure 7c) assumes an anomalous dense body with a fixed, horizontal base and was constrained that the top could not lie above the ground surface. Several base depths were used, with a depth of 3 km providing the best simulation of the observed anomalies. Calculated anomalies are shown in Figure 7d. Parts of the body beneath the central gravity high were assigned a density contrast of 850 kg m^{-3} (corresponding to a density of 3210 kg m^{-3}), which adequately simulated the amplitude of the anomaly. The rest of the body was assigned a density contrast of 540 kg m^{-3} (2900 kg

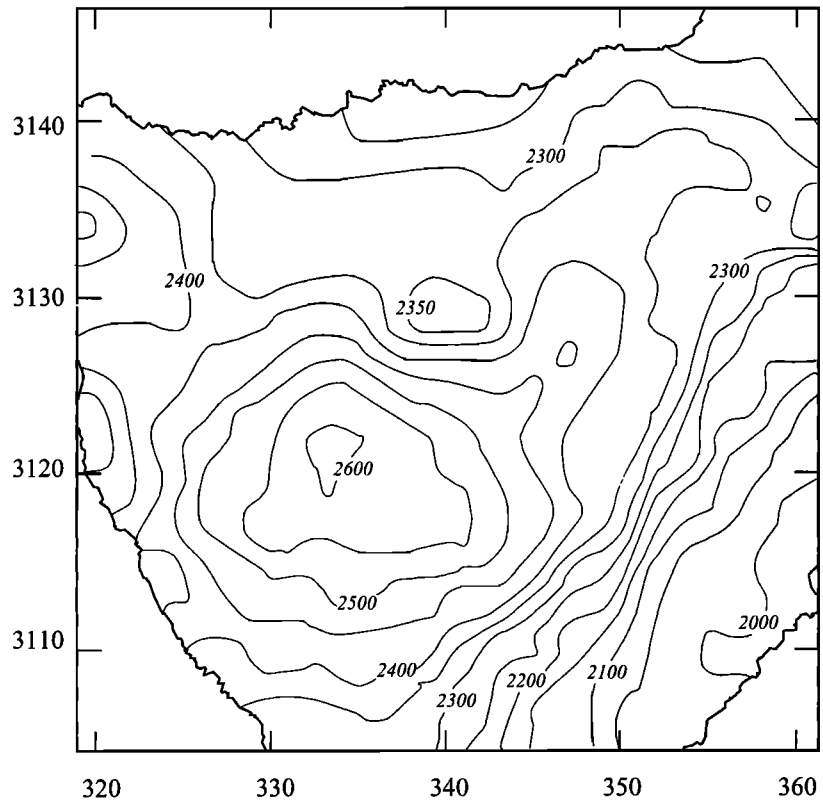

Figure 6. Bouguer anomaly contour map for the central part of Tenerife. Contour interval is 50 gu. The axes correspond to the Spanish National Grid (km UTM).

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Table 1. Density Measurements of Tenerife Lithologies

Sample	Rock Type	Location km UTM X	Location km UTM Y	density, (kg m-3)
S17	old basalt lava	323100	3129300	2340
	(Teno)			2930
				2950
S17	young basalt lava	323100	3129300	2340
	(Teno)			2450
S28	Dorsal Ridge basalt	362250	3124650	1760
				2850
S29	historic basalt	365400	3134250	2890
S8	intermediate lava	346250	3130300	1950
				2180
S12	intermediate lava	335000	3121600	2480
				2170
S13	intermediate lava	333800	3123750	2350
				2110
				2340
S9	phonolitic obsidian	344900	3128700	2250
				2210
				2160
				1990
S10	phonolitic pumice	344025	3127600	1480
	(Montaña Blanca)			1380
				1320
				1360
S11	phonolite lava	340050	3125450	2510
	(Teide-1492 flow)			2160
				2400
S11	phonolitic ignimbrite	335400	3120900	2570
				2510
S24	phonolitic pumice fall	349700	3113350	1100
				1070

m^{-3}). The causative body is centered over the western central part of the island and shows elongation to the NE, beneath the southern part of the Las Cañadas caldera and the Dorsal Ridge, and to the NW, beneath the Teno massif. An important caveat to the interpretation of this model concerns the geometry of the central, high-density subdomain. Owing to the model's simple prismatic structure, this subdomain has the form of a column that intersects the body top at the ground surface. A more realistic geometry would be for the high-density domain to form an inner core to the main part of the body, at depth.

4. Gravity model interpretation

Constraints on the geological plausibility of the gravity models are provided by seismic refraction data for Tenerife [*Watts et al.*, 1997], seismic and gravity studies of other ocean islands [e.g., *Zucca et al.*, 1982], measurements of the densities of volcanic lithologies on Tenerife (Table 1), and estimates of the densities of related plutonic cumulates (Table 2). Analysis of the three-dimensional power spectrum of the gravity anomalies suggests that the tops of the causative structures lie within ~2.6 km of the surface (Figure 8). Strong lateral density contrasts within the island edifice, between the upper parts of Tenerife's volcanic core and its low-density carapace, are inferred to be the main causes of the observed anomalies. This is in agreement with the seismic profile of *Watts et al.* [1997], which shows the presence of a high-velocity core to Tenerife (V_p 5-6 km s^{-1}) surrounded by material of lower seismic velocity (V_p 3-5 km s^{-1}). Of the two end-member models presented, gravity model 2 is considered to be the most geologically plausible, since it has an appropriate geometry to simulate the gravity effects of Tenerife's dense core and shallow, low-density carapace. Model 2 is therefore used to place constraints on the internal structure of the island.

Ocean island cores have densities similar to solid basalt lava because they consist mainly of compacted, massive successions of submarine to subaerial mafic lavas and hypabyssal intrusives, with subordinate plutonic rocks [e.g., *Cendrero*, 1970; *Stillman et al.*, 1975; *Staudigel and Schmincke*, 1984]. The core may thus be modeled as a body with a density equivalent to solid basalt, embedded within a volume corresponding to overlying carapace materials [*Williams and Finn*, 1985]. Carapace materials include diverse volcanic and volcano-sedimentary lithologies containing abundant void space, whose wide density range makes the choice of reduction density problematic. An appropriate reduction density is critical to the model interpretation, since it influences

Table 2. Calculated Densities of Tenerife Plutonic Cumulate Rocks Based on Fractionation Models of *Ablay et al.* [1998]

Mineral Phase	Density, kg m^{-3}	Modelled Mineral Proportions C	D	E	F	G
oliv	4000	45.6	5.0	23.3	0	0
cpx	3400	40.9	58.9	12.1	27.1	11.8
mt	5200	6.4	14.0	4.9	20	7.4
plag	2760	0	19.8	48.5	49	52.4
amph	3300	0	0	0	0	0
afsp	2550	0	0	0	0	0
biot	2700	0	0	0	0	25.8
ilm	4800	6.4	0.6	6.8	0	0
ap	3500	0.6	1.6	3.8	4.7	2.7
Cumulate Density, kg m^{-3}		3770	3430	3140	3280	2870
Rock Type		Plag. Wehrlite to Gabbro		Gabbro to Ti-magnetite Syeno-Gabbro		

Mineral phases used in fractionation models for Tenerife magmas are oliv, olivine; cpx, diopsidic to salitic clinopyroxene; mt, titano-magnetite; plag: plagioclase; amph: kaersutitic amphibole; afsp, alkali feldspar; biot, biotite; ilm, ilmenite; ap, apatite, with mineral densities after *Deer et al.* [1966]. Effects of mineral solid solutions are ignored. The modelled mineral proportions resulting from fractionation calculations for Tenerife magmas are after *Ablay et al.* [1998]. Letters correspond to models given by *Ablay et al.* [1998]. Models C and D refer to fractionation of primitive basanite to evolved basanite, and evolved basanite to plagioclase basanite, respectively. Models E, F, and G refer to fractionation under upper crustal (low pressure) conditions, appropriate for Teide series or Boca Tauce magmas, of plagioclase basanite to phono-tephrite, phono-tephrite to tephri-phonolite and tephri-phonolite to phonolite composition respectively. Cumulate densities and plutonic rock types are also shown.

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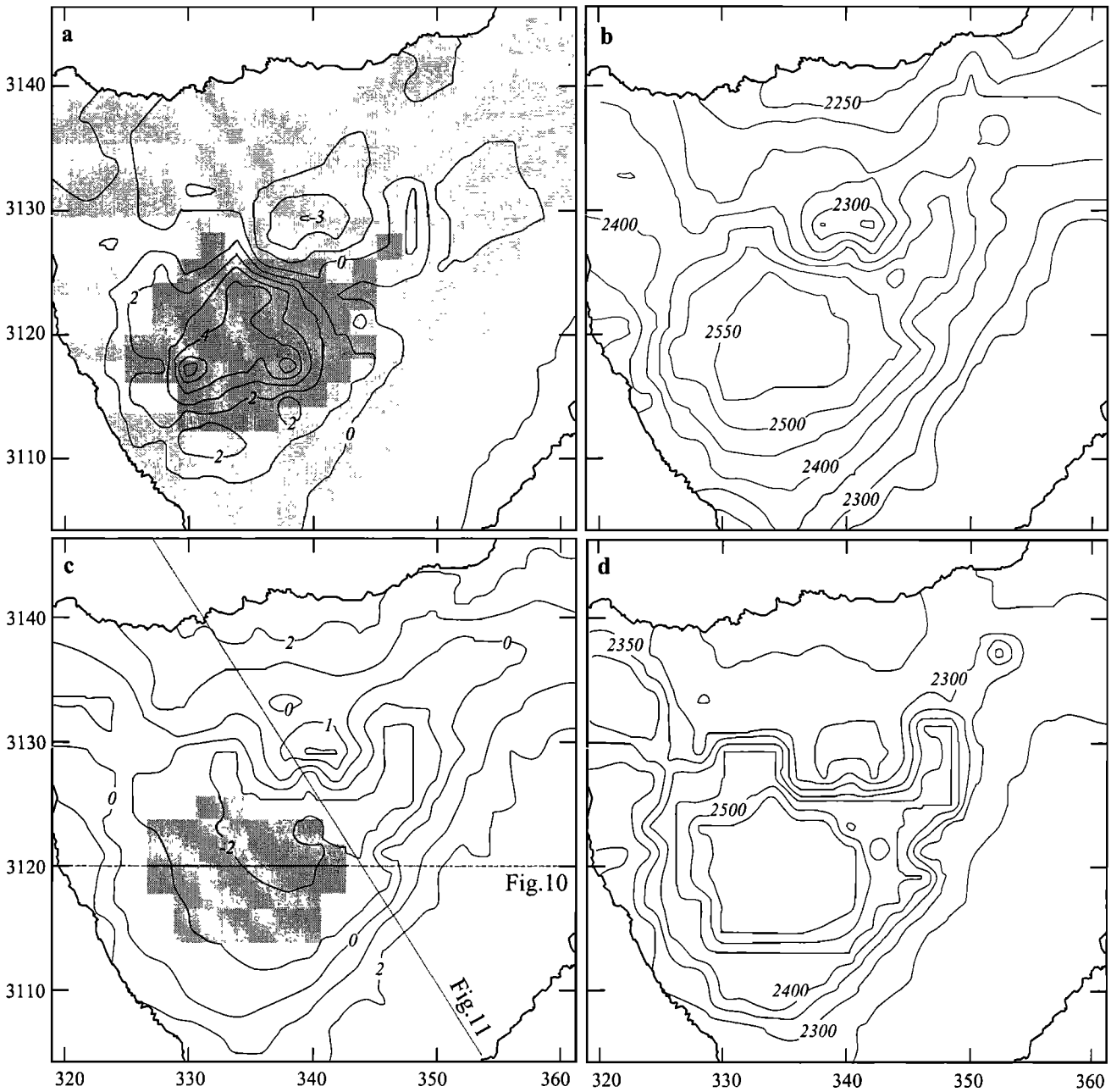

Figure 7. Gravity models and results. (a) Model 1, derived from the Bouguer anomaly map of Figure 6. The model has a fixed top located at a depth of 50 m below the ground surface and is represented by contours on the centers of the bases of the prisms comprising the model, which are 2 km x 2 km in horizontal section. Shading represents the density contrasts of the prisms: dark 840 kg m^{-3}, medium 640 kg m^{-3}, none 190 kg m^{-3}. (b) Computed anomalies of model 1. Contour interval is 50 gu. (c) Model 2, derived from the Bouguer anomaly map of Figure 6. The model has a fixed base located at a depth of 3 km below sea level and is represented by contours on the centers of the tops of the prisms comprising the model, which are 2 km x 2 km in horizontal section. Shading represents the density contrasts of the prisms, dark 850 kg m^{-3}, none 540 kg m^{-3}. The dashed lines correspond to the cross sections shown in Figures 10 and 11. (d) Computed anomalies of model 2. Contour interval is 50 gu.

both the best fit density and the computed geometry of the anomalous body.

Volcanic lithologies making up the shallow country rocks on Tenerife have densities consistent with a mean value of 2360 kg m^{-3} used as the reduction density in the models (Table 1). This value was deduced for central Tenerife (Figure 4) and lies at the upper end of the density range calculated by *Watts et al.* [1997] for the whole island including fringing sediments. For gravity model 2 a density contrast of 540 kg m^{-3} between the anomalous body and surrounding country rocks best simulates the observed anomalies. This equates to a density value of 2900 kg m^{-3} for the main part of the anomalous body, which falls within the upper range of densities measured for massive mafic volcanic rocks from Tenerife (Table 1). The anomalous body computed in gravity model 2 is interpreted to constrain parts of the island made up predominantly of massive mafic lava lithologies such as

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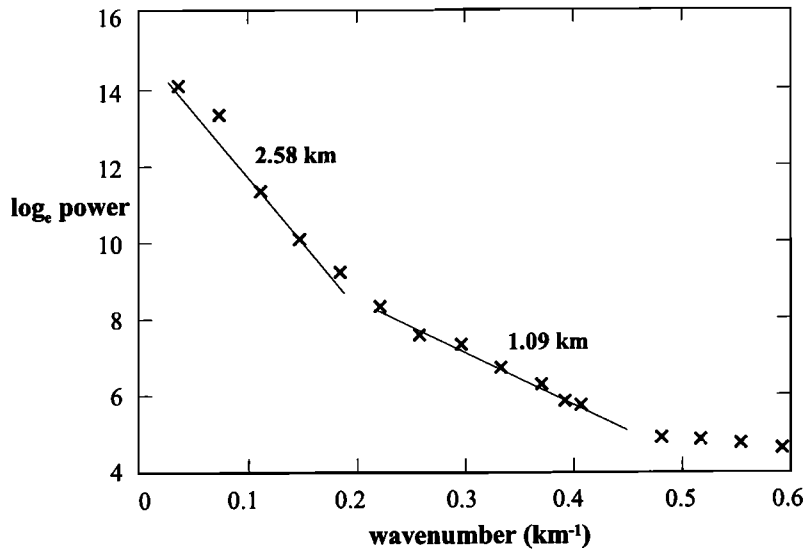

Figure 8. Graph of the natural logarithm of power against wavenumber for the central region of Tenerife containing the Las Cañadas caldera. Values refer to the depth of the sources computed from the linear segments of the power spectrum.

compact successions of mafic lava flows and concentrations of mafic dikes.

Gravity model 2 is compared with surface geological and structural data, complementary seismic and magnetic results [*Watts et al.*, 1997; *Blanco*, 1997] and the previous three-dimensional gravity model of *Camacho et al.* [1991] to provide information about the internal structure of the island. Contours on the top of the body computed in gravity model 2 are shown in Figure 9 with major surface features superimposed for reference. Cross sections through the model are shown in Figures 10 and 11. The model is interpreted in three parts; (1) the SW LCE; (2) the central LCE and Las Cañadas caldera; and (3) postshield rift zones.

4.1. SW Las Cañadas Edifice

A major part of the anomalous body computed in gravity model 2 underlies the SW LCE (Figure 9). In east-west section the body has a subconical form, which is symmetrical about the high-density domain that intersects the surface at Boca Tauce (Figure 10). Exposed locally at Boca Tauce is a compact succession of old, predominantly mafic lavas intruded by abundant mafic dikes. This unit also crops out in deep barrancos on the southern flanks of the LCE and has yielded early Lower Group ages of up to 3.5 Ma for the lavas [*Huertas et al.*, 1994]. Given the geometric caveat described in section 3.3.2., the Boca Tauce unit is interpreted as making up the main part of the anomalous body underlying the SW LCE.

Lavas of the Boca Tauce succession are rich in clinopyroxene, olivine, plagioclase, and Ti-magnetite phenocrysts, and they closely resemble Teide lavas which fractionated at low pressure [*Ablay et al.*, 1998]. Boca Tauce lavas are therefore also interpreted to have evolved within a shallow intrusive system, which provides a possible explanation for the high-density subdomain required beneath Boca Tauce. This sub-domain is given a density of 3210 kg m^{-3}, greater than the main part of the body (2900 kg m^{-3}), in order to simulate the amplitude of the main gravity high on Tenerife (Figure 7c). This value is higher than that measured for any volcanic rock (Table 1) suggesting that other lithologies, not observed at the surface, might contribute to the observed anomalies. Postshield volcanism on Tenerife has involved substantial volumes of differentiated lavas in the compositional range basanite-phonolite [*Ridley*, 1970; *Ablay et al.*, 1998] and intermediate to shallow level plutonic cumulates must therefore be present, as shown by their occurrence as xenolithic nodules in Tenerife lavas and ejecta [e.g., *Borley et al.*, 1971; *Ablay et al.*, 1998]. Table 2 gives density estimates for the main mafic to intermediate plutonic lithologies expected to be found in the core of the island: wehrlite (models C and D) and magnetite gabbros (models E-G). Mineral proportions are derived from the fractionation calculations of *Ablay et al.* [1998] with mineral densities after *Deer et al.* [1966].

Mafic intrusions have been invoked to explain gravity anomalies at Grimsvötn, Iceland [*Gudmunsson and Milsom*, 1997] and Piton de la Fournaise [*Rousset et al.*, 1989]. Grimsvötn is characterized by a 100 gu gravity high which has been modeled in terms of a body of density 2900-3000 kg m m^{-3} at 1.5-4.0 km depth. Piton de la Fournaise exhibits a positive gravity anomaly of 650 gu, which has been modeled as a body with a density contrast predominantly of 600 kg m^{-3}, extending to 6 km depth. The depth range of the causative structures at Grimsvötn and Piton de la Fournaise is comparable with that postulated for Tenerife. However, materials having a density contrast of 840-850 kg m^{-3} with shallow country rocks are required to simulate the amplitude of the Boca Tauce anomaly. This suggests that cumulate rocks may be present, as postulated to explain gravity anomalies over Hawaiian volcanoes of up to 1200 gu [*Zucca et al.*, 1982].

Cumulates expected to be formed during the shallow level fractionation of Tenerife magmas are shown in Table 2. Here, the compositional range of Boca Tauce lavas [see *Fuster et al.*, 1968; *Ridley*, 1970] can be accounted for by a combination of three fractionation steps from evolved basanite through phono-tephrite to tephri-phonolite (Table 2; models D, E, and F). Fractionation across these intervals would be expected to yield a gabbro body with a bulk density of about 3290 kg m^{-3}, sufficient to account for

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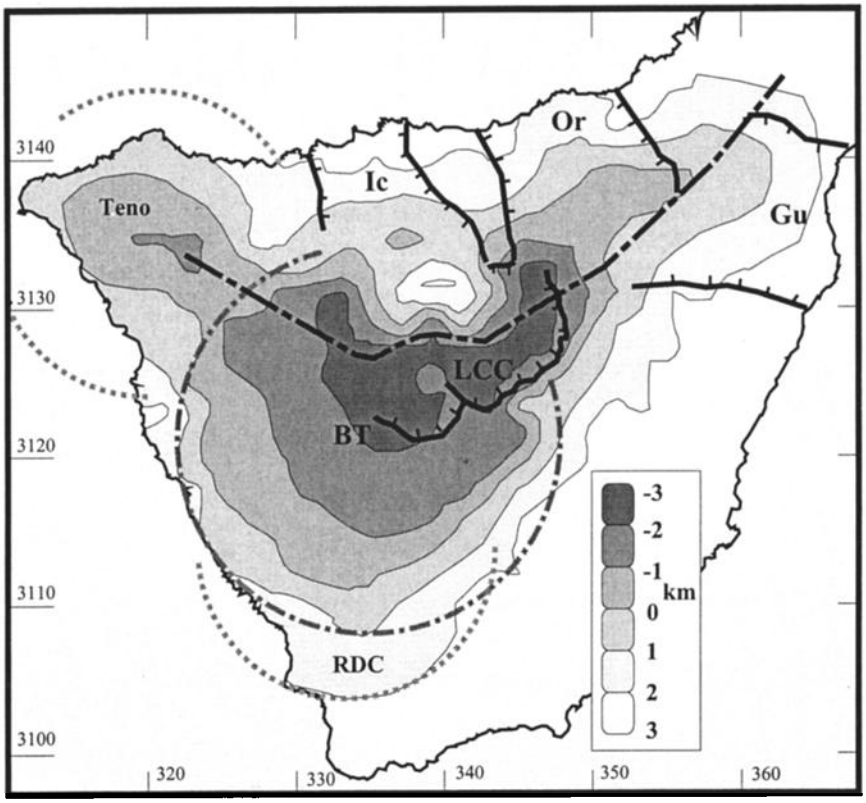

Figure 9. Map showing contours on the surface of gravity model 2, in kilometers below sea level (after Figure 7c) with major surface structural features shown for reference: Thick solid lines with ticks are major scarps: LCC, Las Cañadas caldera walls; Ic, Icod valley; Or, Orotava valley; and Gu, Güimar valley. Thick solid dashed line is axis of the active rift system; thick dashed shaded circles are inferred lateral extents of the Teno, Roque del Conde, and Boca Tauce volcanoes. Circle labeled BT is inferred location of the central vent systems of the Boca Tauce volcano. The axes correspond to the Spanish national grid (km UTM).

the amplitude of the Boca Tauce anomaly. The high densities of the gabbroic cumulates reflect their high titano-magnetite contents (Table 2). This can also explain the nature of the associated magnetic anomaly at Boca Tauce [*Blanco*, 1997], whose high amplitude and negative sign are consistent with the presence of an old, mafic plutonic complex at shallow depth. This mafic plutonic complex is inferred to form the core of an old, predominantly mafic volcano of substantial size, which underlies the SW LCE and is designated the Boca Tauce volcano (Figure 9).

4.2. Central Las Cañadas Edifice

A NW-SE profile through the central LCE (Figure 11) suggests strong N-S asymmetry, as previously suggested by the NE-SW seismic profile of *Watts et al.* [1997]. Beneath the SE flank of the LCE the anomalous body shallows inland and approaches the surface under the southern wall of the Las Cañadas caldera. The body lies at shallow depth beneath the southern caldera floor, whereas beneath the northern caldera and Teide there is a depression in the body occupied by low-density materials. Beneath the NW flank of the LCE, the body lies mainly below sea level. The anomalous body beneath the LCE is interpreted mainly to represent old, compact, mafic volcanics added onto the NE flank of the Boca Tauce volcano. An exception to this is the part of the body underlying the southern caldera floor whose shallow depth suggests materials occupying the caldera (Figure 11). The close spatial correlation between the trace of the caldera wall and the lateral limits of the body top (Figure 9) supports the occupation of the southern part of the caldera by a young, dense caldera fill.

The depression in the body beneath Teide suggests that low-density materials occupy the northern caldera. Using the modified Nettleton's method described above, consideration of isogal maps for progressively lower reduction densities shows that the negative closure over the northern caldera does not disappear until about 1800 kg m^{-3}, suggesting that materials of this particularly low density contribute to the negative anomaly. Such materials are not accounted for in the gravity model so the depth of the low-density region underlying Teide may be exaggerated. The low-density region may define an embayment in the LCE's dense core which is open to the north rather than a closed depression bounded by dense materials on all sides (see Figure 11). *Camacho et al.* [1991] have produced a detailed three-dimensional model of the subcaldera region utilizing a continuously varying density structure which helps resolve this ambiguity. Their model is in good agreement with gravity model 2 and shows the southern caldera to be occupied by high-density materials while the northern caldera is occupied by a low-density region elongated NE-SW and extending down to >3 km below sea level. Their model suggests that very low density materials associated with Teide do contribute to the gravity low over the northern LCE but do not account for it fully, as low-density materials are still required within the northern caldera down to

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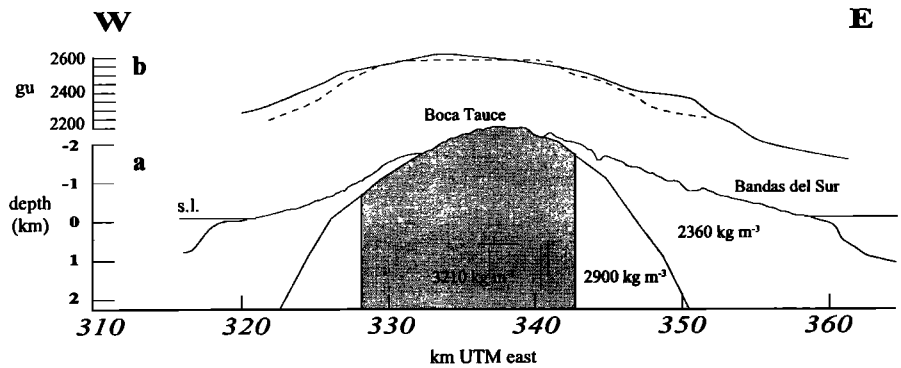

Figure 10. West-east cross section through gravity model 2 at UTM (km north) 3120 (see Figure 7c for location). (a) Model structure. Dark shading is modeled density 3210 kg m^{-3}; light shading is 2900 kg m^{-3}; and unshaded is 2360 kg m^{-3}. Vertical axis is in kilometers below sea level. (b) Comparison of gravity anomalies observed (solid line) and calculated from the model (dashed line).

substantial depths. The presence of a low-density region beneath the northern caldera flanked by dense materials to north and south is also shown, though at a coarser scale, in the seismic profile of *Watts et al.* [1997].

4.3. Postshield Rift Zones

Gravity model 2 suggests that the Santiago and Dorsal ridges comprise dense volcanic cores covered by a significant thickness (1-2 km) of low-density materials (Figure 9). Exposure indicates the low-density cover materials to consist of overlapping scoria cones, mafic lava flows and volcaniclastics. The low bulk density of these materials is attributed to the steep declivity of the rift flank, which favors strong autobrecciation of lava flows and the generation of intercalated volcaniclastic lithologies. The cores of the ridges, which form offshoots to the main part of the anomalous body, are inferred to comprise similar, mafic volcanic lithologies, though highly compacted and heavily intruded by mafic dikes. The absence of an offshoot of the anomalous body beneath the southern volcanic zone (Figure 12) is consistent with it having been a less important axis for intrusion and eruption than the Santiago and Dorsal rifts. Gravity anomalies over the Tenerife rifts do not require the existence of bodies having densities of 3100-3150 kg m^{-3}, such as those interpreted to underlie Hawaiian rifts [*Zucca et al.*, 1982], suggesting that their deep structures differ from those of Hawaiian rift zones.

5. Discussion

5.1. Early Postshield Evolution

The postshield evolution of Tenerife is interpreted to have begun with the growth of a large mafic volcanic edifice, the Boca Tauce volcano. The Boca Tauce volcano is inferred to have been roughly conical and symmetrical about its intrusive core. It is suggested to have overgrown a shield volcano now partly preserved as the Roque del Conde massif and to have overlapped the Teno shield volcano to the NW (Figure 9). Mafic to intermediate lavas at the top of the Boca Tauce sequence have been dated at 3.5 Ma [*Huertas et al.*, 1994], suggesting that the Boca Tauce volcano was built largely before this time. The Boca Tauce volcano is inferred to have contributed to the upper volcanic sequence on the Roque del Conde. This overlies the shield series and has yielded dates of 3.8-3.3 Ma [*Ancochea et al.*, 1990], suggesting that the Boca Tauce volcano was constructed between about 3.8 and 3.5 Ma.

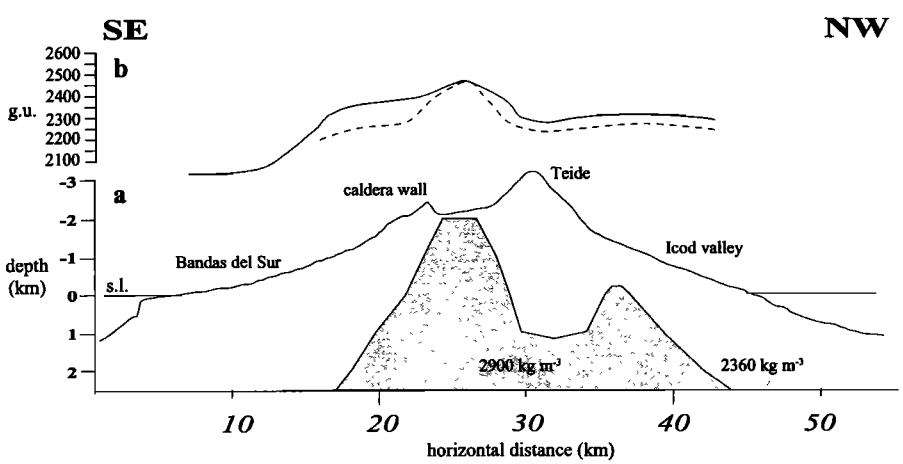

Figure 11. Northwest-southeast cross section through gravity model 2 (see Figure 7c for location). (a) Model structure. Grey shading is 2900 kg m^{-3} and unshaded is 2360 kg m^{-3}. Vertical axis is in kilometers below sea level. (b) Comparison of gravity anomalies observed (solid line) and calculated from the model (dashed line).

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At about 3.5 Ma, mafic volcanism is interpreted to have migrated NE from Boca Tauce to form the LCE and Dorsal Ridge. Mafic lavas dated at 3.3-3.0 Ma are exposed at Boca Tauce and at Las Angosturas (Figure 1) in the southeastern caldera wall [*Martì et al.*, 1994; *Fuster et al.*, 1994; *Ancochea et al.*, 1995], suggesting that the LCE had developed its present elongate form by about 3.0 Ma. Lower Group phonolitic lavas exposed near Boca Tauce at El Cedro (Figure 1), which have yielded ages of up to 3.2 Ma (J. Martí, unpublished data, 1996), are inferred to have been produced by the waning Boca Tauce magma system.

Volcanism is interpreted to have spread NE from the Boca Tauce volcano along a NE-SW lineament, either a rift zone of the Boca Tauce volcano or a preexisting crustal weakness. This lineament is inferred to have since evolved into the active postshield rift system that transects the caldera and extends along the Dorsal Ridge. The later development of the rift system is inferred to have been influenced by the presence of the frozen core of the Boca Tauce volcano, locally preventing extension and limiting the spread of magmatism to the SW. This is inferred to have focused extension onto the Dorsal rift, causing its prolongation to the NE and the later development of the Santiago rift as an accommodatory structure.

5.2. Structure and Evolution of the Las Cañadas Caldera

The development of the LCE after circa 2 Ma involved voluminous phonolitic magmatism thought to have migrated NE accompanied by several vertical caldera collapse events that formed the Las Cañadas caldera [*Araña*, 1971; *Martí et al.*, 1994, 1997]. However, it has also been suggested that the caldera was formed by one or more giant lateral collapses directed toward the north, which have also been linked to the formation of the Icod and Orotava valleys [*Bravo*, 1962; *Navarro and Coello*, 1989; *Coello and Bravo*, 1989]. The identification of >1000 km^3 of chaotic deposits lying off the north coast of Tenerife, which have been interpreted as material removed from the caldera during the formation of the Icod and Orotava valleys, has been cited in support of this [*Watts and Masson*, 1995].

The present work supports a vertical collapse origin for the central Las Cañadas caldera based on the presence of low-density materials 2-3 km below the northern part of the caldera floor. These materials are too deep to be occupying the head of the Icod valley, which is only 0.5-1.0 km deep [see *Ablay and Hurlimann*, 2000]. Indeed, there is no intrinsic reason why the Icod valley should be characterized by a low-density fill. The two other Upper Group lateral collapse valleys, Güimar and La Orotava (Figure 1), show small positive gravity anomalies due presumably to their occupation by young mafic lavas from the Dorsal rift (Figure 9). Nevertheless, a close spatial correlation may be observed between the lateral limits of the low-density region occupying the northern caldera and those of the Icod valley (Figure 9). In view of this, outstanding questions about the origin of the caldera concern the following: (1) the nature of the low-density materials occupying the northern caldera; (2) the nature of the high-density materials occupying the southern caldera; and (3) the role of the Icod landslide in modifying the caldera.

Low-density materials occupying the northern caldera are explained as the collapsed upper parts of the LCE overlain by syncollapse, and postcollapse, pyroclastic and volcaniclastic materials. The upper part of the LCE is downfaulted by about 2 km and juxtaposed against dense materials forming the core of the edifice (Figure 11). The occupation of the southern part of the caldera by dense materials is consistent with the structure of Tenerife's rift system; an axis of mafic intrusion and eruption linking the Santiago rift through the caldera to the Dorsal rift (Figure 9). The intracaldera rift is inferred to have been re-established within the southern (Guajara) caldera after the latter's formation at 0.56 Ma [Martí *et al.*, 1997]. Dense materials occupying the southern caldera are interpreted as rift-related mafic volcanics of Diego Hernandez age (0.56-0.18 Ma) heavily intruded by mafic dyke systems emplaced along the rift axis. The absence of a thick, low-density cover, as inferred to overlie the extracaldera rift flanks, is attributed to ponding of rift products inside the Las Cañadas caldera.

The correlation between the low-density region occupying the northern caldera and the lateral limits of the Icod valley suggests that the Icod landslide modified the northern part of the caldera. The Icod landslide is inferred to have detached from the flank of the intracaldera rift axis, as the Güimar and La Orotava landslides did from the axis of the Dorsal rift. The Icod slide is suggested to have removed the northern caldera walls and part of the dense, rift-related fill from the northern caldera. Teide-Pico Viejo products, erupted after 0.18 Ma, thus rest upon low-density materials in the north of the caldera and dense fill in the south, deepening the negative gravity anomaly over the northern LCC. The southern boundary of the low-density region occupying the northern caldera is interpreted as the buried headwall of the Icod valley. A series of arcuate slope breaks visible in the overlying Teide-Pico Viejo volcanics supports this [*Ablay and Martí*, 2000]. The head of the Icod valley lies directly beneath the active Teide, Pico Viejo, and Montaña Blanca vent systems, which define the present axis of active intracaldera rifting. The buried head of the Icod valley is considered to have played a role in controlling the location of the active volcanoes by focusing extensional stresses.

The inferred location of the Icod valley headwall on the north side of the caldera argues strongly against the Icod landslide having formed the Las Cañadas caldera. The origin of the prodigious volume of chaotic deposits lying off the north coast, previously attributed to the emptying of the caldera [*Watts and Masson*, 1995], therefore requires reconsideration. *Ablay and Hürlimann* [2000] interpret the offshore debris as deriving mainly from a much older episode of mass wasting that affected the northern side of the LCE during Lower Group times. This is consistent with the occurrence in the Tigaiga massif of chaotic deposits overlain by Lower Group lavas dated at 2.2 Ma [*Ibarrola et al.*, 1993]. *Ancochea et al.* [1995] have noted the absence of Lower Group rocks from the north side of the LCE. This is consistent with mass wasting having strongly modified the north side of the LCE as it developed its elongate form between about 3.5 and 2.2 Ma. The early removal of voluminous mafic volcanics from the north side of the LCE helps to explain the depth of the LCE's dense core beneath the north flank of the island.

That the south and SW parts of the LCE do not appear to have been affected by prodigious mass wasting, during either early or late postshield times, whereas the north flank of the LCE and both flanks of the Dorsal Ridge have experienced multiple generations of landslides [*Ablay and Hürlimann*, 2000], is attributed to the buttressing effect of the Boca Tauce volcano. The dense core and buried mafic edifice of the Boca Tauce volcano is inferred to have supported the SW LCE throughout its development, strongly influencing the distribution of portions of unstable flank, and eruptive centres.

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6. Conclusions

1. Gravity anomalies on Tenerife result mainly from lateral density contrasts within the shallow parts of the island edifice between the upper parts of the island's dense core, composed mainly of massive mafic lava lithologies and plutonic rocks, and its overlying carapace of low-density materials.

2. The main gravity anomaly on Tenerife lies over the SW Las Cañadas edifice and represents a buried mafic volcano of substantial size, which is cored by a large mafic plutonic complex. This edifice is interpreted to have developed between circa 3.8 and 3.5 Ma and is designated the Boca Tauce volcano after the inferred location of its central vent systems.

3. The Las Cañadas edifice developed its present dimensions and elongate form by the addition of predominantly mafic volcanic materials onto the NE flank of the Boca Tauce volcano between about 3.5 and 3.0 Ma. Volcanism migrated along a NE-SW lineament, either a rift zone associated with the Boca Tauce volcano or a preexisting structural weakness, which has since evolved into the intracaldera-Dorsal rift system.

4. During Lower Group times (*c*.3 Ma) the Las Cañadas edifice experienced voluminous mass wasting from its northern flanks. This removed mafic material which later would have formed part of the edifice core and produced the majority of the offshore chaotic deposits. The southern flanks of the edifice did not experience major mass wasting due to the buttressing effect of the Boca Tauce volcano.

5. Gravity modeling indicates that the central part of the Las Cañadas caldera is occupied in the south by high-density materials and in the north by low-density materials down to substantial depths. The low-density materials occupy a deep depression in the core of the edifice formed by vertical caldera collapse and probably comprise the foundered upper parts of the LCE plus syncollapse and postcollapse pyroclastics and volcaniclastics. The asymmetry of the preexisting edifice resulted in an asymmetric caldera whose northern wall was lower, and weaker, than that to the south.

6. Following caldera collapse, reestablishment of the intracaldera rift-system led to the occupation of the central caldera by a dense fill. The northern part of the caldera was then modified by a lateral collapse, which formed the Icod valley. The deep negative gravity anomaly over the northern caldera results from low-density Teide-Pico Viejo products covering a low-density caldera fill.

7. The Icod valley landslide detached from the intracaldera rift axis. The headwall of the valley lies inside the central caldera area and does not correlate with the extant walls of the Las Cañadas caldera. The headwall of the Icod valley has focused extensional stresses and controlled the location of the active intracaldera rift and Teide-Pico Viejo vent systems.

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